

FedCND: Federated Graph-Level Clustering under Inter-Client Cluster Number Discrepancy

Junlong Wu*

Hainan University
School of Computer Science and
Technology
Haikou, Hainan, China
ttloung_346@hainanu.edu.cn

Renda Han*

Hainan University
School of Computer Science and
Technology
Haikou, Hainan, China
hanrenda@hainanu.edu.cn

Wenxuan Tu†

Hainan University
School of Computer Science and
Technology
Haikou, Hainan, China
twx@hainanu.edu.cn

Jingxin Liu

Hainan University
School of Cyberspace Security
Haikou, Hainan, China
ljx_0417@hainanu.edu.cn

Haotian Wang

Hainan University
School of Computer Science and
Technology
Haikou, Hainan, China
20010101@hainanu.edu.cn

Jieren Cheng

Hainan University
School of Computer Science and
Technology
Haikou, Hainan, China
992730@hainanu.edu.cn

Abstract

Federated graph-level clustering (FGC) provides an effective solution for analyzing decentralized graph data with privacy protection. Existing methods typically assume that all clients have the same number of clusters. This assumption simplifies the learning task and has achieved preliminary success. However, this assumption rarely holds in practice, as clients often exhibit substantial heterogeneity in both data distributions and semantic granularity. As a result, cluster-specific knowledge becomes misaligned during server-side aggregation, which ultimately degrades the overall clustering performance. To address this challenge, we propose a novel **Federated Graph Clustering under Inter-Client Cluster Number Discrepancy (FedCND)** framework, which aligns inter-client heterogeneous distributions by decoupling graph data into public and private patterns. Specifically, after initial local training and clustering on each client, we design a public learner and a private learner to model public and private graph data, respectively. Only anonymized, cluster-level public information is uploaded to the server, while private information remains local. On the server, cluster-level public prototypes are aggregated based on affinities between reconstructed cluster-level graphs, enabling privacy-preserving prototype alignment across clients with heterogeneous cluster numbers and mitigating interference from misaligned information during global aggregation. Finally, private subgraphs derive client-specific prototypes through local relearning, which are subsequently fused with globally oriented public prototypes for better clustering. Extensive experiments demonstrate that the proposed FedCND achieves an average of 4.9% accuracy improvement against current state-of-the-art methods.

* Both authors contributed equally to this research.

† Corresponding author.



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CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Theory of computation** → *Distributed algorithms*; **Unsupervised learning and clustering**.

Keywords

Federated Learning; Graph-Level Clustering; Prototype Learning

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1 Introduction

The widespread availability of graph-structured data on the Web has motivated Federated Graph Learning (FGL) [23, 24, 32, 33] as a privacy-preserving paradigm for training Graph Neural Networks (GNNs) over decentralized data. FGL enables collaborative learning over distributed graph data while preserving user privacy, making it particularly well suited to web-scale, privacy-sensitive applications such as personalized recommendation [51], decentralized fraud detection [8], and collaborative scientific discovery [25, 56]. Within this context, clustering has emerged as a fundamental task in FGL, driven by the limited availability of labeled data. Recent studies, including FedGCN [27] and FedNCN [28], have made notable progress in advancing federated graph clustering.

Despite recent progress in federated graph clustering, existing methods overlook a critical issue. As illustrated in Fig. 1, they typically assume a uniform number of clusters across clients, i.e., that each client's data is partitioned into the same number of clusters [6]. This assumption simplifies the learning task and theoretical analysis and has led to initial success in federated graph clustering tasks [27]. However, it imposes substantial limitations in practical settings, especially in the presence of heterogeneous graph structures and highly non-independent and identically distributed (non-IID)

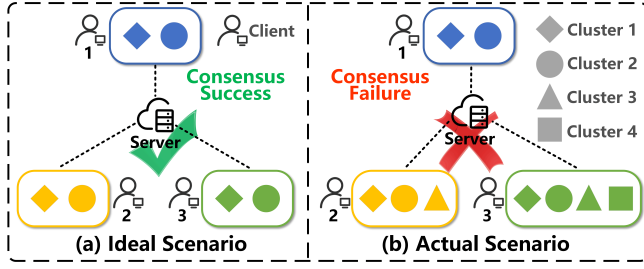


Figure 1: Motivation Illustration. (a) In ideal settings, all clients share the same number of clusters, enabling effective global consensus. (b) In practical scenarios, clients exhibit heterogeneous cluster numbers, resulting in consensus failure.

data [11]. Specifically, 1) this assumption presupposes similar semantic information across clients, whereas real-world graph data often exhibit substantial heterogeneity. Such discrepancies hinder local models from learning accurate semantic partitions, leading to distorted embedding spaces and degraded performance [50]; and 2) forcibly aligning knowledge under heterogeneous requirements induces severe information mismatches during global aggregation, undermining global consistency and cross-client alignment. As a result, both clustering performance and generalization suffer significantly. These limitations highlight the need for a robust and flexible federated graph clustering framework that preserves client-specific structures while enabling stable aggregation under heterogeneous cluster counts.

A natural solution is to transmit cluster-level signals to the server and aggregate cluster-wise information across clients. To achieve this, however, two key challenges require to be well addressed: 1) how to effectively extract shared structures within clusters under heterogeneous graph distributions, while preventing the leakage of sensitive information during client-to-server communication; and 2) how to efficiently align cluster-level prototypes from different clients to maintain global consistency across training rounds without sacrificing local specificity. To address the first challenge, we decouple cluster-level data into public and private patterns, sharing only anonymized public information while retaining private information locally. This design distills essential cluster characteristics for prototype alignment while suppressing interference from data heterogeneity and sensitive information during aggregation. To address the second challenge, we propose a dual-prototype collaborative optimization mechanism that iteratively integrates globally oriented public prototypes with locally oriented private prototypes, jointly enhancing global consistency and local personalization.

Based on the above observations, we propose FedCND, a novel unsupervised federated graph-level clustering framework. The core idea of FedCND is to decouple common and personalized patterns from graph data. Specifically, on each client, local graph instances are first subjected to initial representation learning and clustering. For graph data within each cluster, a public-private subgraph decoupler is applied to partition the data into public and private subgraphs. Correspondingly, a public learner and a private learner are employed to learn representations from the public and private

graph data, respectively. On the server side, a model consistent with the local public learner is used to implicitly recover clustering features from noisy subgraph representations and reconstruct cluster-level graphs in a privacy-preserving manner. Graph kernels are then employed to evaluate affinities between cluster-level graphs across clients, enabling dynamic alignment of public prototypes. The updated global prototypes are then broadcast to all clients. Each client subsequently fuses these global prototypes with private prototypes obtained through local relearning to form unified representations, which guide the next round of clustering. Our main contributions are summarized as follows:

- To the best of our knowledge, this work is the first to explicitly investigate the problem of inter-client cluster number discrepancy in federated graph clustering.
- We propose **Federated Graph Clustering under Inter-Client Cluster Number Discrepancy (FedCND)**, a novel framework that effectively mitigates distribution shifts and semantic misalignment caused by heterogeneous semantic granularity across clients.
- Extensive experiments have been conducted on twenty unlabeled, cross-dataset and cross-domain non-IID graph benchmarks, demonstrating the effectiveness and superiority of **FedCND** over state-of-the-art methods.

2 Related Work

2.1 Federated Graph Clustering

Federated Graph Learning (FGL) enables collaborative GNN training under strict privacy constraints. Within this domain, Federated Graph Clustering (FGC) focuses on unsupervised discovery of group structures across distributed clients. Initial efforts utilize parameter aggregation or prototype sharing to mitigate structural heterogeneity. For instance, FedGCN [27] generates global consensus representations via multi-source prototype sharing and Gaussian estimation. To address structural fragmentation in partitioned graphs, FedNCN [28] recovers cross-subgraph links by integrating clustering priors with a joint MLP-GNN optimization framework. Recent research focuses on data imperfections and model generalization. To handle incomplete information, FedAI [15] utilizes path digests and graph kernels for attribute inference, while IFedGC [31] employs causally-aware global structures to guide feature completion. To further address non-IID distributions, FedIGL [49] introduces invariant learning and bi-gradient regularization to decouple client-specific patterns from invariant structures, mitigating spurious correlations. Furthermore, PERFECT [30] facilitates personalized clustering through privacy-preserving representative sampling and min-max optimization. Our work differs from previous FGL studies in that we focus on FGC under inter-client cluster number discrepancy. We leverage disentangled prototype learning and graph kernel-based affinity to align semantic information without requiring a uniform number of clusters across all clients.

2.2 Inter-Client Cluster Number Discrepancy

Addressing the discrepancies in cluster numbers across heterogeneous clients poses a fundamental challenge in unsupervised federated learning. Several recent works have attempted to relax the assumption of a fixed, uniform cluster count. For instance, FKM

[12] extends K -means to the federated setting, specifically addressing the convergence challenges that arise when the number of clusters varies between local centers. To accommodate dynamic environments, eFedG [37] proposes an evolving Gaussian clustering method for data streams; it utilizes a merge-and-add mechanism to adjust cluster quantities in real-time without relying on a pre-defined global. Similarly, FedCRef [36] employs a reconstruction error-based refinement strategy to uncover global data distributions in multi-cluster-per-client scenarios, allowing clients to iteratively discover distinct clusters without label supervision. Despite these advancements, existing methods primarily focus on Euclidean data or specific streaming modalities and may struggle to disentangle shared global patterns from client-specific variations when cluster numbers differ significantly. Direct aggregation in such heterogeneous settings often leads to the misalignment of cluster-specific knowledge. In contrast, our proposed FedCND addresses these limitations within the domain of graph-level clustering. By explicitly decoupling graph data into public and private subgraphs, FedCND aligns heterogeneous distributions via anonymized public prototypes while retaining unique private patterns locally. This design effectively mitigates the interference caused by misaligned information during global aggregation, ensuring robust clustering performance even under substantial inter-client cluster number discrepancies.

2.3 Prototype Learning

Prototype learning serves as a cornerstone in representation learning and clustering, where latent centroids capture cluster-level semantics. Prototypical Networks [39] pioneered this by representing classes as embedding means to facilitate few-shot classification, a paradigm subsequently extended to metric learning [20] and domain adaptation [38]. In the graph domain, prototypes aggregate high-level semantics across nodes or subgraphs. For example, Graph Prototypical Networks (GPN) [10] utilize representative neighborhood vectors for few-shot classification, while PGCL [26] leverages prototype alignment to regularize unsupervised graph representations. These approaches establish prototypes as essential anchors for disentangling common and task-specific graph structures. Recent advancements have integrated prototypes into federated learning. FedProto [41] communicates class prototypes rather than full model weights to optimize bandwidth and privacy. However, its reliance on homogeneous label distributions limits performance in non-IID scenarios. In contrast, our work introduces client-side adaptive clustering with variable cluster counts. By decoupling public and private subgraphs and employing cluster-level prototype learning, we simultaneously enable global knowledge sharing and robust privacy preservation.

3 Methodology

Fig. 2 shows the architecture of the FedCND. The core idea is to separate public patterns and private patterns within the same cluster. The public patterns are uploaded to the global server for information aggregation, while the private patterns are retained locally for further learning. This process mainly includes three stages: public-private subgraph decoupling, global cluster-level prototype alignment and dual-prototype fusion.

3.1 Notations

We consider a federated setting with M clients. The total number of clusters across all clients is given by $O = \sum_{m=1}^M C_m$, where C_m denotes the number of clusters held by client m . For simplicity, we take a single client as an example and consider other clients similarly: each client holds a set of private and distinct graph datasets with C_m clusters, where I denotes the total number of graphs held by that client, denoted as $\mathcal{G} = \{G_1, G_2, \dots, G_I\}$. For each G_i , the node feature matrix is denoted by $\mathbf{X}_i \in \mathbb{R}^{N \times d}$, and the original adjacency matrix by $\mathbf{A}_i \in \mathbb{R}^{N \times N}$, where N is the number of nodes in the G_i and d is the dimension of the node features.

3.2 Local Clustering Model Warm-Up

This section introduces local model learning and clustering, aiming to partition the graph data into clusters on the client side, thereby laying the foundation for subsequent cluster-level public-private subgraph decoupling operations. We first employ a Graph Isomorphism Network (GIN) to encode the structural and feature information of each graph. Given a graph $G_i = (\mathbf{A}_i, \mathbf{X}_i)$, the node embeddings $\mathbf{H}_i \in \mathbb{R}^{N \times d'}$ are computed through GIN:

$$\mathbf{H}_i^{(l)} = \text{MLP}^{(l)} \left((1 + \epsilon) \cdot \mathbf{H}_i^{(l-1)} + \mathbf{A}_i \cdot \mathbf{H}_i^{(l-1)} \right), \quad (1)$$

where $\mathbf{H}_i^{(0)} = \mathbf{X}_i$, d' denotes the dimensionality of the node embedding, ϵ is a learnable parameter, and $\text{MLP}^{(l)}$ denotes the multi-layer perceptron at the l -th layer. To optimize the learned node embeddings $\mathbf{H}_i$, we minimize the following composite objective function:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \mathcal{L}_{\text{KL}}, \quad (2)$$

where $\mathcal{L}_{\text{MSE}}$ denotes the reconstruction loss, and $\mathcal{L}_{\text{KL}}$ represents the clustering loss, formulated in accordance with FedGCN [27].

To obtain a holistic representation for each graph, we apply a readout function over the final node embeddings to compute the graph-level embedding $\mathbf{z}_i$ as follows:

$$\mathbf{z}_i = \text{READOUT}(\mathbf{H}_i) = \sum_{v \in G_i} \mathbf{h}_v, \quad (3)$$

where $\mathbf{h}_v$ is the embedding of node v in graph G_i . Given the I graph-level embeddings $\mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_I] \in \mathbb{R}^{I \times d'}$, on a client, we apply K -means to partition them into C_m clusters. The local graph dataset $\mathcal{G}$ is then divided into C_m disjoint subsets $\{\mathcal{G}_1, \dots, \mathcal{G}_{C_m}\}$.

In summary, the proposed local clustering model warm-Up offers two major advantages: it 1) learns the local graph representation space, thereby establishing a reliable foundation for clustering, and 2) partitions client-level graph data into cluster-level graph representations, thereby establishing a principled foundation for subsequent cluster-level public-private subgraph decoupling.

3.3 Public-Private Subgraph Decoupling

After clustering the I graphs of a client into C_m clusters, we introduce a public-private subgraph decoupling mechanism to handle heterogeneous graph structures. Each cluster's graphs $\mathcal{G}_c = \{G_i\}_{i=1}^{|\mathcal{G}_c|}$ are split into public and private subgraphs: public subgraph capture common patterns and are shared with the server for global knowledge aggregation, while private subgraphs remain on the client to enhance personalized representations.

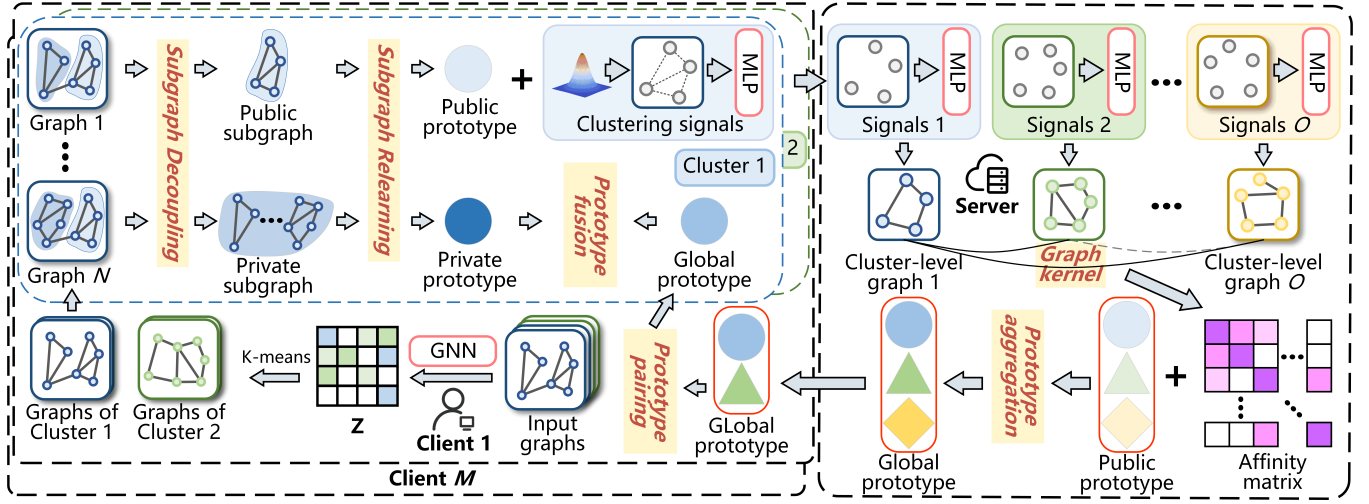


Figure 2: The framework of FedCND. Specifically, each client first clusters its local graph data and then decouples the graph data belonging to each cluster into public and private subgraphs. Public subgraphs are perturbed with Gaussian noise to extract clustering signals, from which the server reconstructs cluster-level graphs and computes their affinities via a graph kernel to align public prototypes. Updated global prototypes are then sent back to clients, where they are fused with locally retained private prototypes to guide the next round of clustering.

First, to capture the most frequent subgraph structure of each cluster, we propose the Adaptive Frequent Subgraph Discovery (AFSD) mechanism. Unlike traditional frequent subgraph discovery [21] which relies on a rigid, pre-defined support threshold, AFSD adaptively identifies the most representative substructures by formulating the discovery process as a stability-driven optimization problem. This design avoids the need in traditional methods to manually set different support thresholds for each cluster when processing graph data from different clusters.

AFSD lies in its ability to dynamically adjust the selection criteria based on the structural distribution within each cluster. Specifically, for the c -th cluster of the m -th client, AFSD seeks a public subgraph S_c^{pub} that maximizes a stability measure S^{sta} , which quantifies the consistency of a substructure across heterogeneous graph samples:

$$S_c^{\text{pub}} = \arg \max_{S^{\text{sta}}} \phi(S^{\text{sta}} | G_1, G_2, \dots, G_{|\mathcal{G}_c|}). \quad (4)$$

To achieve this without manual thresholding, we define the stability S^{sta} as a function of the subgraph's occurrence and its distribution across the cluster:

$$\phi(S^{\text{sta}} | \mathcal{G}_c) = \frac{1}{|\mathcal{G}_c|} \sum_{i=1}^{|\mathcal{G}_c|} \exp(-\text{Var}(S^{\text{sta}}, G_i)), \quad (5)$$

where $\text{Var}(S^{\text{sta}}, G_i)$ represents the structural discrepancy of S^{sta} within the cluster. Next, for each graph $G_i \in \mathcal{G}_c$, we define its private subgraph S_i^{pri} as the remaining structure after removing the extracted public subgraph S_c^{pub} :

$$S_i^{\text{pri}} = G_i \setminus S_c^{\text{pub}}, \quad \forall G_i \in \mathcal{G}_c. \quad (6)$$

Here, $\{S_i^{\text{pri}}\}_{i=1}^{|\mathcal{G}_c|}$ is retained locally and not transmitted to the server. We then feed the S_c^{pub} and $\{S_i^{\text{pri}}\}_{i=1}^{|\mathcal{G}_c|}$ into dual-branch learners: a public subgraph learner $\mathcal{F}^{\text{pub}}$ and a private subgraph learner

$\mathcal{F}^{\text{pri}}$. The public learner produces the embedding of the public subgraph, while the private learner first produces embeddings for each private subgraph, which are then aggregated to form a single private prototype. Formally, we have:

$$\mathbf{P}_c^{\text{pub}} = \mathcal{F}^{\text{pub}}(S_c^{\text{pub}}), \quad \mathbf{P}_c^{\text{pri}} = \frac{1}{|\mathcal{G}_c|} \sum_{i=1}^{|\mathcal{G}_c|} \mathcal{F}^{\text{pri}}(S_i^{\text{pri}}), \quad (7)$$

where $\mathbf{P}_c^{\text{pub}}$ represents the public prototype capturing the shared patterns across the cluster, and $\mathbf{P}_c^{\text{pri}}$ represents the private prototype. The private prototype is retained locally, and after the public prototypes are aggregated globally, the updated public prototypes are sent back to the local to be fused with the private prototype.

To upload the node features of the public subgraph from the client to the server while preserving privacy, we follow the masking mechanism of FedNCN [28] and generate a random Gaussian noise matrix $\mathbf{M}^{(c)} \in \mathbb{R}^{n_c \times d}$ with the same shape as $\mathbf{X}_c^{\text{pub}} \in \mathbb{R}^{n_c \times d}$, where n_c denotes the number of nodes in the public subgraph.

We then locally employ a multi-layer perceptron (MLP) f_θ as a projector, and extract clustering signals $\mathbf{G}_{\text{MLP}}^{(c)}$ from the gradients generated during MLP training:

$$\mathbf{G}_{\text{MLP}}^{(c)} = \nabla_\theta \mathcal{L}_{f_\theta}(\mathbf{X}_c^{\text{pub}} - \mathbf{M}^{(c)}), \quad (8)$$

where θ denotes the MLP parameters, ∇_θ denotes the gradient operator with respect to the MLP parameters, and $\mathcal{L}_{f_\theta}(\mathbf{X}_c^{\text{pub}} - \mathbf{M}^{(c)})$ is the loss function used to train the MLP. By uploading $\mathbf{G}_{\text{MLP}}^{(c)}$, $\mathbf{M}^{(c)}$ and $\mathbf{P}_c^{\text{pub}}$ to the server, the server can obtain informative clustering signals for the public subgraph S_c^{pub} while avoiding direct exposure of sensitive client data $\mathbf{X}_c^{\text{pub}}$ [28].

In summary, the proposed public-private subgraph decoupling offers two major advantages: it 1) retains private subgraph patterns

locally, enhancing personalized representations on each client and mitigating interference from heterogeneous or sensitive data during global aggregation, and 2) enables the extraction of clustering signals in a privacy-preserving manner, avoiding privacy leakage caused by directly uploading raw data.

3.4 Global Cluster-level Prototype Alignment

To handle inconsistent and heterogeneous local clusters in federated graph clustering, we propose a global prototype alignment mechanism. It reconstructs cluster-level graphs, measures inter-cluster affinities in a latent space, and performs structure-aware aggregation to derive global prototypes.

Using a global MLP f_{θ_g} , we recover node features $\tilde{\mathbf{X}}_c^{\text{pub}} \in \mathbb{R}^{n_c \times d}$ from the O aggregated cluster signals $\mathbf{G}_{\text{MLP}}^{(c)}$ and noise masks $\mathbf{M}^{(c)}$. This process, based on FedNCN [28], reconstructs feature matrices $\mathbf{X}_c$ by matching their gradient behavior against the global model: $\tilde{\mathbf{X}}_c^{\text{pub}} = \arg \min_{\mathbf{X}_c} \|\nabla_{\theta_g} \mathcal{L}(f_{\theta_g}(\mathbf{X}_c, \mathbf{M}^{(c)})) - \mathbf{G}_{\text{MLP}}^{(c)}\|_2^2$. This formulation reconstructs node features $\tilde{\mathbf{X}}_c^{\text{pub}}$ that preserve local update semantic information while protecting raw private attributes.

After obtaining $\tilde{\mathbf{X}}_c^{\text{pub}}$, we embed these features into a low-dimensional manifold to preserve local geometric relationships: $\tilde{\mathbf{H}}_c = \Phi_{\text{embed}}(\tilde{\mathbf{X}}_c^{\text{pub}})$, where $\tilde{\mathbf{H}}_c \in \mathbb{R}^{n_c \times d'}$, and $\Phi_{\text{embed}}(\cdot)$ denotes an embedding function. We construct cluster-level graphs $\tilde{\mathcal{S}}_c = (V_c, E_c)$, where edges in E_c are established via feature proximity among n_c nodes. The adjacency matrix $\tilde{\mathbf{A}}_c \in \mathbb{R}^{n_c \times n_c}$ is defined as:

$$\tilde{\mathbf{A}}_c(i, j) = \exp\left(-\|\tilde{\mathbf{H}}_c(i, \cdot) - \tilde{\mathbf{H}}_c(j, \cdot)\|_2^2\right), \quad v_i, v_j \in V_c. \quad (9)$$

We obtain O cluster-level graphs $\{\tilde{\mathcal{S}}_c\}_{c=1}^O$ capturing public structural regularities. To align cross-client clusters, an affinity matrix $\mathbf{K} \in \mathbb{R}^{O \times O}$ is computed using a normalized Weisfeiler-Lehman (WL) graph kernel [17]:

$$k_{ij} = \text{GK}(\tilde{\mathcal{S}}_i, \tilde{\mathcal{S}}_j). \quad (10)$$

Finally, the global prototype $\mathbf{P}_c^{\text{glo}}$ is obtained by a weighted aggregation of public prototypes: $\mathbf{P}_c^{\text{glo}} = \sum_{j=1}^O \frac{k_{cj}}{\sum_{t=1}^O k_{ct}} \mathbf{P}_j^{\text{pub}}$. The resulting global prototypes are then distributed to clients and fused with their private prototypes.

This stage provides three key benefits: 1) it ensures privacy by reconstructing node attributes from signals and masks without exposing raw data; 2) it aligns cross-client semantic information via affinity-based fusion; and 3) it enhances global consistency and supports local dual-prototype integration.

3.5 Dual-Prototype Fusion

The dual-prototype fusion stage is designed to integrate global knowledge with client-specific patterns. After the globally aligned prototypes $\{\mathbf{P}_c^{\text{glo}}\}_{c=1}^{C_m}$ are returned to each client, we further enhance the guiding function by fusing them with private prototypes $\{\mathbf{P}_c^{\text{pri}}\}_{c=1}^{C_m}$. Here, C_m denotes the number of clusters for each client. Specifically, we first compute the distance between each pair of prototypes $\mathbf{P}_c^{\text{glo}}$ and $\mathbf{P}_{c'}^{\text{pri}}$ to build the Euclidean distance matrix $\mathbf{D} \in \mathbb{R}^{C_m \times C_m}$, as follows:

$$d_{c,c'} = \|\mathbf{P}_c^{\text{glo}} - \mathbf{P}_{c'}^{\text{pri}}\|_2. \quad (11)$$

We utilize $\mathbf{D}$ to find the optimal one-to-one matching $\pi: [\mathbf{P}_c^{\text{glo}}] \rightarrow [\mathbf{P}_c^{\text{pri}}]$ that minimizes the total distance. We solve:

$$\pi^* = \arg \min_{\pi \in T_{C_m}} \sum_{c=1}^{C_m} \mathbf{D}_{c, \pi(c)}, \quad (12)$$

where T_{C_m} is the set of all permutations on $\{1, \dots, C_m\}$; this is efficiently computed via the Hungarian algorithm [7]. Since the number of prototypes is much smaller than the number of graph samples, applying this algorithm for prototype matching does not incur significant computational overhead. Once matched, each pair is fused via a mixing coefficient $\alpha \in [0, 1]$:

$$\mathbf{P}_c^{\text{fus}} = \alpha \mathbf{P}_c^{\text{glo}} + (1 - \alpha) \mathbf{P}_{\pi^*(c)}^{\text{pri}}, \quad (13)$$

resulting in the client's fused prototypes $\{\mathbf{P}_c^{\text{fus}}\}_{c=1}^{C_m}$. The hyperparameter α , which controls the relative influence of the global versus personalized components, is further examined through hyperparameter sensitivity analysis experiments. The resulting fused prototypes $\{\mathbf{P}_c^{\text{fus}}\}_{c=1}^{C_m}$ are subsequently employed to steer the client's model optimization and clustering process. Through this dual-prototype fusion mechanism, the framework achieves a favorable balance between global coherence and local personalization.

4 Experiments

4.1 Experiment Settings

Benchmark Datasets. We conduct an evaluation of FedCND across a range of widely public datasets from the TUDataset collection, encompassing diverse domains such as small molecules (MUTAG, BZR_MD, PTC_MR, BZR, COX2, DHFR, AIDS), bioinformatics (ENZYMES, DD, PROTEINS), synthetic (SYNTHETIC, Synthie), computer vision (MSRC_9, Letter-low, Letter-med, Letter-high, MSRC_21, MSRC_21C), and social networks (COLLAB, IMDB).

Evaluation Metrics. To comprehensively assess the performance of FedCND, we adopt four standard unsupervised clustering evaluation metrics: Accuracy (ACC) [43–46, 48, 53], Normalized Mutual Information (NMI) [14, 24, 32], Adjusted Rand Index (ARI) [3, 13, 29], and F1 Score [34, 42, 47].

Baseline Methods. To rigorously evaluate the effectiveness of FedCND, we select advanced FGC methods, i.e., FedGCN [27], and other FGL methods to adapt them to FGC, such as FedSage [55], GCFL [52], FedStar [40], FGAD [5], and LG-FGAD [4].

Implementation Details. All methods are implemented with PyTorch and run on a single NVIDIA GeForce RTX 3090 GPU. Each client uses a three-layer GIN [54] for graph-level embedding, with batch size 256 and learning rate 0.001. We employ Adam [19] as the optimizer for both the local and global models. The FL process runs for 10 communication rounds, with 10 training epochs per round on both client and server.

4.2 Unsupervised Comparison Experiments

4.2.1 Under Cluster Number Discrepancy. To validate the superiority of FedCND, we construct six non-IID settings, where each client holds an independent dataset. The number of clients is determined following the settings of FedGCN [27], rather than limiting it to only a small number of clients. FedSage, GCFL, and FedStar, originally supervised FGL methods, are adapted into unsupervised clustering

Table 1: Performance comparison of FGC tasks under inter-client cluster number discrepancy. Notably, all compared methods are evaluated under an unsupervised scenario on six non-IID settings to ensure a fair comparison. The number in parentheses (top-right) of each setting denotes the number of clusters in client data. * denotes that these supervised FGL methods are adapted to the unsupervised scenario. † denotes that these unsupervised FGL tasks are manually converted into FGC tasks.

Models	MN ^(2,3)				MNY ^(2,3,4)				MNYB ^(2,3,4,6)			
	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1
FedSage* (NeurIPS 21)	43.8±1.5	8.8±1.2	10.7±1.4	25.5±2.2	40.6±2.4	9.2±1.5	10.3±1.4	22.9±3.8	37.2±1.1	12.3±1.6	14.6±0.9	25.6±2.3
GCFL* (NeurIPS 21)	48.9±2.1	2.0±1.5	0.1±0.2	27.7±1.4	44.0±1.1	4.0±1.0	0.4±0.4	36.8±1.1	42.0±1.1	4.5±1.0	0.6±0.2	30.8±2.7
FedStar* (AAAI 23)	49.1±0.9	2.0±0.7	1.7±0.6	30.3±1.3	44.1±1.3	3.7±1.4	0.8±0.5	34.8±2.1	42.3±1.7	4.4±2.0	2.5±2.1	35.6±0.9
FGAD† (ACM MM 24)	54.0±0.7	11.3±0.7	10.4±1.0	42.7±0.9	56.7±1.2	20.1±0.4	19.5±0.6	53.4±1.7	49.8±0.3	19.7±0.7	16.1±1.0	46.5±1.0
LG-FGAD† (IJCAI 24)	51.3±1.1	12.9±2.4	5.7±1.5	45.0±1.2	57.2±1.9	20.4±1.4	15.2±1.9	53.0±2.2	52.0±2.3	21.9±1.2	12.3±1.8	48.2±2.8
FedGCN (AAAI 25)	55.5±1.2	12.6±2.5	12.3±3.1	41.8±2.7	61.2±0.7	16.2±4.2	18.5±4.3	49.5±3.0	57.4±0.4	15.3±3.1	18.1±2.8	47.6±1.4
FedCND (Ours)	61.1±0.7	20.7±0.4	18.7±0.4	49.3±2.4	63.9±0.6	24.1±1.5	28.0±3.1	57.3±2.3	60.7±1.0	25.2±1.9	27.8±1.7	54.5±1.3
Models	MNYBC ^(2,3,4,6,8)				MNYBCV ^(2,3,4,6,8,15,20)				CV ^(15,20)			
	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1
FedSage* (NeurIPS 21)	36.4±1.6	11.1±2.2	13.9±1.5	20.8±3.3	32.0±1.4	8.7±0.3	12.5±1.2	18.6±2.7	24.1±1.6	20.3±1.3	10.3±4.2	21.9±1.9
GCFL* (NeurIPS 21)	43.5±1.3	7.6±1.0	3.9±1.3	35.9±2.5	35.4±1.5	13.9±1.1	6.6±1.2	27.0±1.9	30.8±2.6	31.9±1.1	17.4±1.1	29.8±1.7
FedStar* (AAAI 23)	43.9±1.1	7.4±1.6	3.5±1.4	37.3±2.2	36.1±1.2	13.6±1.0	6.0±1.8	31.0±1.3	31.4±1.7	31.4±1.5	16.7±1.5	30.3±1.7
FGAD† (ACM MM 24)	46.0±0.8	18.1±0.7	12.6±0.9	43.5±1.6	42.0±0.6	29.7±0.9	12.7±1.6	38.8±1.0	27.6±1.0	38.1±0.7	7.2±0.8	25.9±1.1
LG-FGAD† (IJCAI 24)	51.8±1.6	21.7±0.7	12.2±0.9	48.1±2.0	48.7±1.5	29.5±2.1	21.2±2.2	45.3±1.7	36.7±1.2	41.1±0.9	17.9±0.8	31.3±1.4
FedGCN (AAAI 25)	54.4±4.9	12.6±5.1	13.0±6.7	44.7±2.9	44.6±0.7	17.5±3.4	12.8±4.2	34.8±2.7	31.8±2.5	34.2±2.2	17.0±2.5	28.4±2.7
FedCND (Ours)	60.7±0.8	24.7±1.1	27.1±1.1	54.3±1.1	54.1±0.6	35.0±1.1	29.7±1.2	48.6±1.0	42.9±1.1	48.6±0.1	30.3±0.9	40.5±1.0

Table 2: Performance comparison of FGC tasks under inter-client cluster number consistency. The number in parentheses (top-right) of each setting denotes the number of clusters in client data. * denotes that these supervised FGL methods are adapted to the unsupervised scenario. † denotes that these unsupervised FGL tasks are manually converted into FGC tasks.

Models	SM ²				SM-SY ²				SM-BIO ²			
	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1
FedSage* (NeurIPS 21)	55.6±1.4	12.2±1.3	7.6±0.6	50.2±1.0	49.4±1.7	10.4±1.5	11.2±2.3	42.6±2.1	57.4±2.2	5.2±2.1	4.2±2.7	49.9±0.5
GCFL* (NeurIPS 21)	61.1±1.8	8.7±2.4	9.4±2.4	43.3±1.6	52.0±0.9	4.3±2.1	0.8±1.1	49.6±2.3	60.1±1.8	4.7±2.4	3.2±2.3	47.3±1.5
FedStar* (AAAI 23)	58.9±2.4	12.0±1.2	0.1±0.8	49.7±2.8	53.1±2.6	3.9±1.3	0.2±0.1	41.2±2.4	59.5±1.6	5.3±1.5	3.8±2.0	51.7±2.2
FGAD† (ACM MM 24)	65.8±0.8	18.8±1.9	3.4±1.1	62.9±0.6	60.8±1.9	10.1±3.3	7.0±3.4	59.3±2.2	59.6±1.5	9.0±1.4	7.8±1.7	56.0±2.6
LG-FGAD† (IJCAI 24)	66.4±2.4	20.2±2.6	4.3±3.2	63.8±2.6	65.5±1.5	15.2±2.3	11.4±2.0	63.8±1.9	63.5±1.0	14.7±1.1	2.1±2.0	60.7±1.5
FedGCN (AAAI 25)	75.9±0.8	23.1±1.6	31.1±3.4	67.1±1.5	70.8±2.5	14.2±7.0	16.9±8.5	58.9±5.6	69.2±0.6	14.0±2.7	17.5±3.1	59.1±0.9
FedCND (Ours)	76.1±0.4	24.3±0.3	32.1±0.5	68.0±0.7	71.3±1.4	16.1±2.7	17.3±2.1	64.1±2.7	70.7±1.3	15.1±1.6	18.0±1.7	60.9±2.3
Models	SM-BIO-SY ²				SN ²				CV ¹⁵			
	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1
FedSage* (NeurIPS 21)	57.6±1.9	20.6±1.9	17.6±2.4	46.7±1.8	53.3±1.9	14.8±1.4	11.6±2.8	49.3±2.0	10.1±1.4	30.5±1.7	13.6±1.8	10.4±1.7
GCFL* (NeurIPS 21)	59.1±2.0	14.4±2.2	13.7±2.8	52.3±1.9	52.1±2.3	12.5±2.3	13.2±2.3	52.3±1.6	13.7±2.0	17.7±2.4	14.3±2.7	13.2±1.4
FedStar* (AAAI 23)	57.9±2.6	15.7±2.4	16.1±3.0	52.3±2.2	51.7±2.7	13.7±2.8	12.4±1.9	50.7±2.3	12.4±2.7	22.4±2.5	15.3±2.1	11.6±1.9
FGAD† (ACM MM 24)	58.4±0.5	7.6±0.4	6.4±0.8	54.6±0.7	61.2±1.1	18.4±2.0	12.7±1.6	47.1±1.3	28.9±1.6	21.5±1.8	12.0±1.4	26.7±1.9
LG-FGAD† (IJCAI 24)	62.2±1.3	14.6±2.6	3.0±2.9	56.7±0.8	64.0±1.4	22.1±2.0	15.3±1.9	49.7±1.0	31.2±1.2	24.1±1.5	14.2±1.6	29.8±2.0
FedGCN (AAAI 25)	68.6±1.3	13.5±2.1	17.2±3.6	59.4±3.8	66.6±2.3	30.4±6.6	34.1±5.3	50.7±2.4	34.6±2.8	34.2±1.4	19.3±1.8	31.6±3.1
FedCND (Ours)	70.1±1.6	16.7±1.0	18.1±1.3	60.4±2.2	67.0±1.3	30.9±2.3	35.0±2.3	53.1±1.6	35.7±1.2	35.6±1.6	21.3±0.6	32.1±0.6

versions following the approach of FedGCN [27]. Although FGAD and LG-FGAD are unsupervised FGL methods, they are designed for graph anomaly detection tasks. We modify their task to focus on federated graph-level clustering. In contrast, FedGCN and FedCND are specifically designed for federated graph-level clustering. The experimental results are shown in Table 1. Here, the MN non-IID setting contains the datasets DHFR, AIDS, COLLAB, and IMDB-MULTI; the MNY non-IID setting contains DHFR, AIDS, IMDB-MULTI, and SYNTHIE; the MNYB non-IID setting contains DHFR, AIDS, IMDB-MULTI, SYNTHIE, and ENZYMES; the MNYBC non-IID setting contains DHFR, AIDS, IMDB-MULTI, SYNTHIE, ENZYMES, and MSRC_9; the MNYBCV non-IID setting contains Letter-med, AIDS, IMDB-MULTI, SYNTHIE, ENZYMES, MSRC_9, and MSRC_21C; and the CV non-IID setting contains Letter-low, Letter-med, Letter-high, MSRC_21, and MSRC_21C. Three key observations can be drawn

from the results: 1) FedSage, GCFL, FedStar, FGAD, and LG-FGAD are not designed for federated graph-level clustering, and thus their effectiveness is limited in this context. 2) Notably, FedCND even outperforms FedGCN, which is tailored for federated graph-level clustering tasks. This can be attributed to FedCND’s ability to aggregate knowledge globally at the cluster-level, which adapts to heterogeneous data distributions arising from different numbers of clusters across clients. This enhances the model’s generalization across heterogeneous cluster counts among clients. 3) Only the public patterns are globally aggregated, while private patterns are retained locally, ensuring that global clustering relies solely on high-quality clustering information free from locally specific noise.

4.2.2 Under Cluster Number Consistency. To validate the clustering performance of FedCND under uniform cluster counts, we adopt the experimental setting proposed in FedGCN [27] as a benchmark for

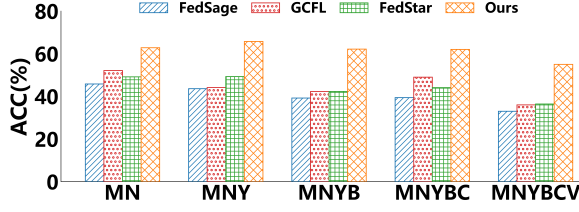


Figure 3: Performance comparison between three advanced federated graph-level classification methods and FedCND under five non-IID settings.

assessment. The details of the non-IID setting are identical to those in FedGCN. As shown in Table 2, the experimental results allow us to draw the following three conclusions: 1) by decoupling public and private patterns through a public-private subgraph decoupler, private patterns are retained locally within each client, thereby enhancing the quality of client-specific representations. 2) the dual-prototype collaborative optimization mechanism utilizes public prototypes to locally compensate for private prototypes, effectively reducing the distributional discrepancy between global and local data. 3) global information aggregation is conducted at the cluster-level rather than the client-level, enabling finer-grained semantic information.

4.3 Comparison with Supervised Methods

To further validate the effectiveness of FedCND, we compare it with three supervised methods trained on few labeled samples. The few-shot labeling setup follows the configuration of FedGCN [27]. As shown in Fig. 3, our method consistently outperforms or remains competitive with these supervised baselines, despite relying solely on unlabeled data. This performance gain comes from our model’s disentanglement of public and private patterns, enabling global clustering to rely on clean, interference-free signals while enhancing local personalized representations.

4.4 Ablation Study

To analyze the contribution of each component in FedCND, we conduct an ablation study by progressively removing or modifying key modules and measuring their impact on clustering performance. Specifically, we evaluate the following six ablation variants: (1) **Local**: Learning is performed solely based on local models without any collaboration. (2) **w/o Subgraph Decoupler (SD)**: The subgraph decoupler is disabled, and each client directly uploads the original prototypes to the server for affinity computation. (3) **w/o public Prototype (CP)**: The dual-prototype collaboration mechanism is removed. Only the global public prototypes are used to optimize the local model. (4) **w/o Private Prototype (PP)**: The dual-prototype collaboration mechanism is removed. Only the local private prototypes are used for model optimization. (5) **w/ Alternative Aggregation**: Replace the global affinity-based aggregation method with alternative aggregation strategies (e.g., FedProx [22], FedPer [1], FedAvg [35]). (6) **w/ Alternative Graph Kernels**: The Weisfeiler-Lehman (WL) graph kernel [17] used in this work is replaced with other graph kernels, including the Random Walk (RW) kernel [18], Lovász Theta (LT) kernel [16], Cosine (COS) kernel [9],

and Shortest Path (SP) kernel [2]. The performance results of these variants are summarized in Table 3. From the table, we can observe the following key findings: 1) The subgraph decoupler and dual-prototype fusion mechanism play a crucial role in enhancing both local personalization and global consistency. 2) In settings with minimal inter-client cluster disparity (e.g., MN and CV), retaining local private prototypes yields better performance; whereas in settings with large disparities (e.g., MNYBC and MNYBCV), global public prototypes perform better. This demonstrates that the dual-prototype mechanism enhances local optimization when client distributions are similar and emphasizes global consistency when they differ significantly, thereby achieving synergistic optimization. 3) FedCND outperforms FedAvg, FedPer, and FedProx by leveraging graph kernel-based affinity computation and cluster-aware global aggregation, enabling it to adaptively aggregate more beneficial information for each client. 4) The WL (Weisfeiler-Lehman) kernel used in FedCND performs best, superior to COS, LT, RW, and SP kernels, due to its stronger ability to capture local structural variations, resulting in more accurate affinity estimation and improved clustering performance.

4.5 Hyperparameter Sensitivity Analysis

To investigate the impact of the global public prototype and local private prototype in Eq. 13 on model performance, we conduct a sensitivity analysis by varying the weighting hyperparameter α . As shown in Fig. 4, the results reveal that FedCND benefits more from local private prototypes when client cluster distributions are similar, while global public prototypes contribute more significantly when cluster differences among clients are large. This demonstrates that both types of prototypes are crucial for improving clustering performance, and their joint optimization effectively enhances the model.

4.6 Convergence Analysis

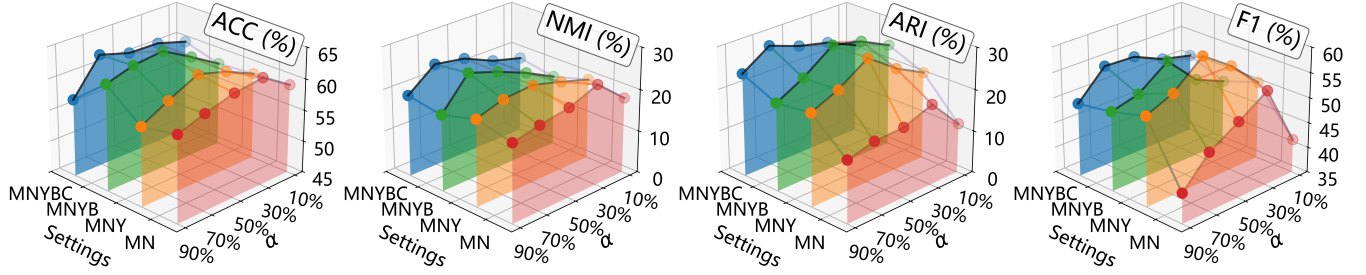
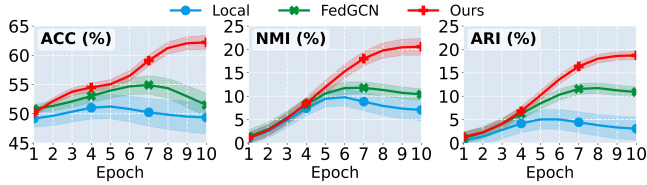
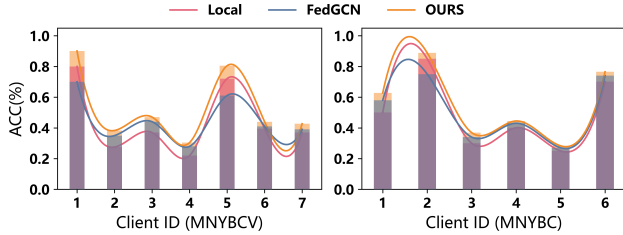
To evaluate the stability and convergence behavior of our proposed FedCND, we monitor the training dynamics of local training, FedGCN, and FedCND across communication rounds under the MNYBC non-IID setting. As shown in Fig. 5, our model exhibits smooth and monotonic convergence across all objectives. This indicates that FedCND demonstrates strong stability and numerical controllability during the federated optimization process, effectively reducing training error through iterations and progressively approaching the optimal solution. Furthermore, the smooth convergence trend of the model reflects its robustness to fluctuations in gradient estimation, along with high optimization efficiency and convergence speed, validating that our approach maintains consistency and robustness even under non-IID federated settings.

4.7 Clustering Performance of Each Client

To evaluate the clustering performance at the client level in FedCND, we compare the clustering performance of individual clients under MNYBCV and MNYBC non-IID settings, contrasting three baselines: training only on local data (Local), FedGCN, and our method (FedCND). Results shown in Fig 6 indicate that FedGCN can actually perform worse than the Local baseline on several clients: we attribute this to negative transfer during traditional

Table 3: Ablation study of FedCND under four non-IID settings. The number in parentheses (top-right) of each setting denotes the number of clusters in client data.

Models	MN ^(2,3)			MNYBC ^(2,3,4,6,8)			MNYBCV ^(2,3,4,6,8,15,20)			CV ^(15,20)		
	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
Local	50.2±2.5	4.2±2.0	3.1±1.5	50.5±3.0	10.0±2.5	10.5±2.0	45.0±2.8	18.0±3.0	13.0±2.5	30.0±3.0	34.0±5.0	18.0±4.0
w/o SD	53.4±2.0	7.0±4.6	5.1±2.5	53.0±3.4	13.5±3.2	13.7±4.4	47.4±3.1	23.6±4.7	18.3±5.1	32.6±3.6	36.8±5.5	20.7±4.1
w/o CP	57.1±1.4	16.8±1.6	13.2±2.2	55.6±3.1	16.9±3.0	17.6±4.6	50.5±2.9	28.5±2.8	23.7±3.6	38.8±2.4	44.5±2.8	26.6±2.8
w/o PP	57.5±2.1	14.4±2.4	11.5±2.8	56.6±3.0	18.1±2.8	19.1±2.9	51.7±2.5	30.2±2.1	25.0±2.1	37.4±2.3	41.5±2.8	24.8±2.8
FedProx	56.4±0.5	13.8±1.7	12.4±2.4	56.6±0.7	20.9±2.3	22.5±2.6	50.3±0.5	31.5±1.2	26.5±1.3	38.5±1.2	43.5±0.2	27.7±0.5
FedPer	56.7±0.8	13.5±0.4	12.0±0.8	56.3±0.6	20.7±2.6	23.1±2.2	50.0±0.6	30.9±0.6	25.6±0.7	38.1±1.2	43.6±0.1	27.7±0.4
FedAvg	56.4±0.6	13.4±0.5	12.1±0.7	56.8±0.7	20.6±2.5	22.9±2.4	49.9±0.7	31.0±0.8	26.2±0.9	38.0±1.0	45.1±0.2	27.6±0.5
COS kernel	56.4±0.8	15.4±0.6	13.2±0.5	55.0±0.9	17.0±1.0	19.9±0.9	49.4±1.0	31.1±1.2	23.1±1.3	38.1±1.2	41.9±0.3	23.0±0.7
LT kernel	57.1±1.0	16.0±0.7	13.0±0.6	54.7±1.0	18.7±1.2	21.1±1.0	47.8±0.9	29.0±1.1	23.0±1.2	35.9±1.3	41.6±0.4	24.0±0.6
RW kernel	59.3±0.7	16.4±0.5	16.0±0.6	58.1±0.8	20.6±1.0	24.3±1.2	50.6±0.6	<u>34.2±0.9</u>	<u>27.8±1.0</u>	37.1±1.0	<u>48.0±0.2</u>	<u>29.6±0.8</u>
SP kernel	59.9±0.5	18.1±0.4	16.4±0.5	58.9±0.6	22.7±0.9	24.9±1.0	53.7±0.7	31.8±1.0	27.3±1.2	41.2±1.0	45.7±0.2	27.7±0.7
Ours	61.1±0.7	20.7±0.4	18.7±0.4	60.7±0.8	24.7±1.1	27.1±1.1	54.1±0.6	35.0±1.1	29.7±1.2	42.9±1.1	48.6±0.1	30.3±0.9

**Figure 4: The sensitivity of FedCND with the variation of the hyperparameter α under four non-IID settings.****Figure 5: Convergence study under the MNYBC non-IID setting, ACC, NMI, and ARI employed as evaluation metrics.****Figure 6: Comparison of ACC across clients on the MNYBCV and MNYBC non-IID setting.**

global aggregation when clients have different numbers of clusters and varying semantic granularity. In such cases, cluster-related information becomes misaligned at the server, so aggregated updates

overwrite useful local structure and degrade client-level clustering. By contrast, FedCND improves clustering on almost every client by explicitly separating public and private patterns, while performing cluster-level prototype alignment exclusively on anonymized public patterns. By keeping client-specific data local, FedCND minimizes the noise introduced by heterogeneous and private data during the global aggregation process.

5 Conclusion

We propose FedCND, a federated graph-level clustering framework that tackles the challenge of varying cluster numbers across clients. A public-private subgraph decoupler is designed to separate public and private patterns, uploading only cluster-level clustering signals to ensure privacy and improve global clustering quality. Private prototypes are kept locally to enhance client-specific representations. A dual-prototype fusion mechanism further aligns global prototypes via graph kernel matching and integrates them with local prototypes, jointly optimizing global consistency and local personalization.

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